EHR Forecast

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This project predicts first-recorded diagnoses within five years from each patient's past synthetic EHR. The final model uses 512 features and 40 LightGBM classifiers. Official-validation mAP is **0.246334** and macro AUROC is **0.762251**; hidden-test scores are unavailable.

Official split	Patients	Use
Train	2,791	Fit preprocessing, feature selection and models
Validation	365	Score the fixed model; no fitting
Test	358	Generate probabilities; outcomes withheld

Data and features

One input is a patient's records strictly before the anchor. For train and validation, the anchor is five years before the last encounter (STOP, or START if STOP is missing); test anchors are supplied. Labels mark a condition's first-ever diagnosis from the anchor through five calendar years later, inclusive. Previously diagnosed targets receive probability zero.

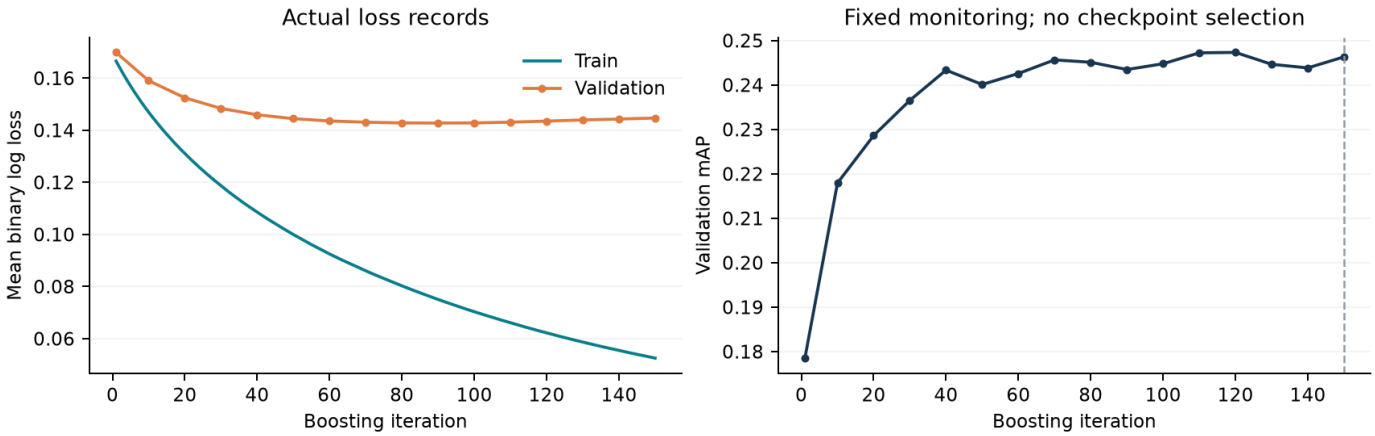
Candidate features combine age, gender, race, ethnicity and birthplace with ten structured tables: diagnoses, encounters, medication starts, procedures, observations, allergies, care plans, devices, immunizations and imaging examinations. Counts and recency summarize history; measurements retain units and missingness, with summaries of level, variability and change. Categorical histories, visit links and clinical interactions add context.

Only valid intervals ending before the anchor contribute durations; other ends are treated as unavailable. Annual QALY/DALY/QOLS, costs and reason fields are excluded. DNA, image pixels and ECG payloads lack adequate temporal provenance and are also excluded. Structured imaging facts remain eligible. No external data or pretrained weights are used.

Training-only quality checks retain 16,181 of 36,327 candidate columns. Mutual information, shallow-tree gain and sparse logistic coefficients select 512. Three 80% training subsamples provide a separate stability check. The final set includes five explicitly imaging-related features and none explicitly ECG-related.

Final model and training

Each target has a binary cross-entropy model with 150 trees, at most 31 leaves per tree, learning rate 0.05, minimum leaf size 20, max_bin 127, L2=20, and 0.8 row/column sampling. Seeds are 20261016 plus target index. The final fit used all 2,791 training patients, with no early stopping or ensemble.



Training loss comes from the actual fit. Validation curves were replayed after the model and predictions were fixed; they were not used to choose a checkpoint. The final model uses all 150 trees.

Model comparison

The historical comparison used three-fold cross-validation on 1,675 fitting patients, followed by scoring on 558 development patients, all from official train. Preprocessing and selection were refitted per fold. Ten model families, two feature controls and three blends were compared; confirmation outcomes were not used for tuning.

Model or control	OOF mAP	Dev mAP	Dev AUROC
LightGBM (512)	0.1728	0.2408	0.7590
LightGBM (earlier strict features)	0.1707	0.2349	0.7578
Nonnegative blend	0.1800	0.2331	0.7684
LightGBM (all quality columns)	0.1714	0.2330	0.7599
Equal top-three blend	0.1760	0.2273	0.7607
Equal ten-model blend	0.1732	0.2165	0.7694
CatBoost	0.1696	0.2112	0.7531
XGBoost	0.1649	0.2096	0.7505
TabM	0.1613	0.1976	0.7483
Elastic-net logistic	0.1620	0.1904	0.7458
ExtraTrees	0.1433	0.1814	0.7287
Random forest	0.1391	0.1810	0.7329
EBM	0.1380	0.1781	0.7359
MLP	0.1437	0.1754	0.7342
RBF SVM	0.1432	0.1748	0.7066

LightGBM had the highest development mAP; no blend improved that point estimate. The gain over earlier strict features was 0.00590, with a paired patient-bootstrap 95% interval [-0.00607, 0.01925] conditional on fixed models. The interval includes zero. Development data were reused, and OOF (out-of-fold) scores informed selection; these are not independent test scores.

Results and limitations

Prostate targets ranked highest: AP 0.972 with 11 positives and 0.938 with eight. PSA-related history is a plausible signal. Pneumonia and obesity also ranked well. Wrist sprain and foot laceration were weak; sparse events and limited predictive history may explain part of the gap.

Final training loss was 0.05254 versus 0.14470 on raw validation probabilities. This suggests overfitting, but does not isolate its contribution from population differences or sampling noise. Official validation was reused historically, and former internal holdouts entered the final fit. Results from one seed and synthetic data do not establish performance on new clinical populations.

AI workflow and reproducibility

I used Codex in a repeated cycle of planning, review, implementation and verification. I first asked it to propose a staged plan, then reviewed the approach and intermediate results, questioned assumptions and refined the next steps before further implementation.

Codex assisted with schema inspection, leakage audits, feature engineering, model comparisons, tests and documentation. PLAN.md tracked scope and dependencies. Development logs recorded changes, rationale, evidence, limitations and next steps. Failed checks led to investigation and revision before retesting; saved configurations and W&B records linked experiments to their results.

Before release, all 39 synthetic tests passed in a fresh Python 3.12 environment on Windows. The published code and Hub model reproduced both prediction files and all validation metrics exactly.

Code: github.com/LL333111/ehr-forecast-final
Model: huggingface.co/X13579/ehr-forecast-final
W&B: cloud training | fixed-model validation replay.

Data: Coherent Synthetic Data Set, generated by Synthea. Copyright 2022 The MITRE Corporation, Public Release Case 21-1917; CC BY 4.0. Raw data is not redistributed.

Appendix: validation by condition

365 patients; Pos. = new diagnoses. AP and AUROC are shown for all 40 targets.

Code	Condition	Pos.	AP	AUROC
444814009	Viral sinusitis	99	0.415	0.751
195662009	Acute viral pharyngitis	72	0.232	0.625
10509002	Acute bronchitis	61	0.296	0.667
88805009	Chronic congestive heart failure	29	0.339	0.824
230690007	Stroke	35	0.239	0.738
22298006	Myocardial Infarction	19	0.722	0.926
68496003	Polyp of colon	18	0.156	0.845
44465007	Sprain of ankle	16	0.046	0.507
53741008	Coronary Heart Disease	15	0.100	0.714
233604007	Pneumonia	17	0.623	0.983
43878008	Streptococcal sore throat	22	0.084	0.525
65966004	Fracture of forearm	16	0.111	0.664
126906006	Neoplasm of prostate	11	0.972	0.999
271737000	Anemia	15	0.126	0.794
64859006	Osteoporosis	12	0.088	0.811
443165006	Pathological fracture due to osteoporosis	14	0.305	0.890
49436004	Atrial Fibrillation	14	0.088	0.778
92691004	Carcinoma in situ of prostate	8	0.938	0.998
36971009	Sinusitis	9	0.043	0.561
75498004	Acute bacterial sinusitis	5	0.046	0.755
162573006	Suspected lung cancer	10	0.155	0.841
424132000	Non-small cell carcinoma of lung TNM stage 1	9	0.124	0.830
254637007	Non-small cell lung cancer	9	0.125	0.830
39848009	Whiplash injury to neck	8	0.028	0.513
26929004	Alzheimer's disease	13	0.116	0.849
263102004	Fracture subluxation of wrist	10	0.054	0.563
72892002	Normal pregnancy	10	0.604	0.991
713197008	Recurrent rectal polyp	10	0.223	0.899
62106007	Concussion with no loss of consciousness	7	0.091	0.725
15777000	Prediabetes	10	0.122	0.832
58150001	Fracture of clavicle	8	0.168	0.570
162864005	Body mass index 30+ - obesity	11	0.627	0.984
16114001	Fracture of ankle	8	0.176	0.700
70704007	Sprain of wrist	8	0.020	0.411
283371005	Laceration of forearm	7	0.040	0.526
314994000	Metastasis from malignant tumor of prostate	6	0.642	0.993
284551006	Laceration of foot	7	0.017	0.388
22325002	Abnormal gait	11	0.156	0.859
723857007	Silent micro-hemorrhage of brain	11	0.166	0.861
156073000	Fetus with unknown complication	5	0.230	0.971